

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3.	(a)			0.8 (1) Conversion to 800 kW (1) 6 hrs (1) $800 \times 6 = 4800$ (1) ecf can be allowed for incorrect data taken from the graph as long as working shown Answer of 4.8×10^n on answer line – 3 marks where n is not 3	1	3		4	4	
	(b)	(i)		$3600/0.6 = 6000$ (1) $6000 \times 3 = 18\,000$ (1)		2		2	2	
		(ii)		6000 (1) 6000×0.55 (1) = 3300 km ² (1) Allow: answer for part (i) $\times 0.55$ (1) = ans (1) e.g. $18\,000 \times 0.55$ (1) = 9900 (1)		3		3	3	
	(c)	(i)		Cost of generating electricity per kW/hr is 2.8p with nuclear compared with 5.6 p for wind/ nuclear unit is half the price of wind (1) Need to replace wind turbine 3 times / Need 18 000 turbines for each nuclear power station / accept you need 6000 turbines (1) £54 000 million compared £4000 million for one nuclear power station (over lifetime) (1) No numerical values – 0 marks			3	3		

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		(ii)		No waste by wind compared to <u>radioactive</u> waste in nuclear (1) { <u>more</u> land required / <u>more</u> habitats destroyed/ <u>more</u> visual pollution} for the equivalent number of wind turbines (1)			2	2		
	(d)			Carbon footprint is much <u>greater</u> in a coal fired power station (1) As it (continually) <u>burns</u> coal (which releases CO ₂) (1)	2			2		
				Question 3 total	3	8	5	16	9	0